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#include <stdio.h>
#include "platform.h"
#include "xil_printf.h"
#include "xscugic.h"
#include "xparameters.h"
#include "xscutimer.h"
#include "xil_exception.h"

XScuTimer_Config *tC;
XScuTimer tI;

XScuGic_Config *iC;
XScuGic iI;

void tmrHandler(){
    XScuTimer_DisableInterrupt(&tI);

    if(XScuTimer_IsExpired(&tI))
    {
        xil_printf("We have an Interrupt at 1 Sec\n");
    }

    XScuTimer_ClearInterruptStatus(&tI);
    XScuTimer_EnableInterrupt(&tI);
    XScuTimer_RestartTimer(&tI);
}

void tmrInit(){
    int status = 0;
    tC = XScuTimer_LookupConfig(XPAR_PS7_SCUTIMER_0_DEVICE_ID);
    status = XScuTimer_CfgInitialize(&tI,tC,tC->BaseAddr);
    if(status == XST_SUCCESS)
        xil_printf("Timer Init Successful\n");
    else
        xil_printf("Timer Init Failed\n");

    XScuTimer_Stop(&tI);
    XScuTimer_LoadTimer(&tI,333333333); ///// 1 sec
    XScuTimer_DisableAutoReload(&tI);
}

void intrInit(){
    int status =0 ;
    iC = XScuGic_LookupConfig(XPAR_SCUGIC_0_DEVICE_ID);
    status = XScuGic_CfgInitialize(&iI, iC, iC->CpuBaseAddress);
    if(status == XST_SUCCESS)
        xil_printf("GIC Init Successful\n");
    else
        xil_printf("GIC Init Failed\n");

    Xil_ExceptionInit();
    Xil_ExceptionRegisterHandler(XIL_EXCEPTION_ID_IRQ_INT, (Xil_ExceptionHandler) XScuGic_InterruptHandler, &iI);
    Xil_ExceptionEnable();

    XScuGic_Connect(&iI, XPAR_SCUTIMER_INTR, (Xil_InterruptHandler)tmrHandler, &tI);
    XScuGic_Enable(&iI, XPAR_SCUTIMER_INTR);

    XScuTimer_EnableInterrupt(&tI);
}

int main()
{
    init_platform();
    tmrInit();
    intrInit();

    XScuTimer_Start(&tI);

    while(1);

    cleanup_platform();
    return 0;
}

```